

Effect of the fat composition of a single high-fat meal on inflammatory markers in healthy young women.

The aim of the present study was to examine the effect of a single high-fat meal with different fat quality on circulating inflammatory markers and gene expression in peripheral blood mononuclear cells (PBMC) to elucidate the role of fat quality on postprandial inflammation. A postprandial study with fourteen healthy females consuming three test meals with different fat quality was performed. Test days were separated by 2 weeks. Fasting and postprandial blood samples at 3 and 6 h after intake were analysed. The test meal consisted of three cakes enriched with coconut fat (43 % energy as saturated fat and 1 % energy as α -linolenic acid (ALA)), linseed oil (14 % energy as ALA and 30 % energy as saturated fat) and cod liver oil (5 % energy as EPA and DHA and 5 % energy as ALA in addition to 31 % energy as saturated fat). In addition, ex vivo PBMC experiments were performed in eight healthy subjects investigating the effects of EPA and ALA on release and gene expression of inflammatory markers. The IL-8 mRNA level was significantly increased after intake of the cod liver oil cake at 6 h compared with fasting level, which was significantly different from the effect observed after the intake of linseed cake. In contrast, no effect was seen on circulating level of IL-8. In addition, ALA and EPA were shown to elicit different effects on the release and mRNA expression levels of inflammatory markers in PBMC cultured ex vivo, with EPA having the most prominent pro-inflammatory potential.